

FINAL DOCUMENTATION

UNIFIED MILITARY ANALYTICS AND COMPARISON DASHBOARD

The Unified Military Analytics and Comparison Dashboard is a comprehensive data analytics project designed to analyze and compare global military power across 140+ countries.

The system integrates collecting data from GlobalFirepower.com, data engineering, KPI feature development, and interactive dashboard visualization into a unified analytics ecosystem.

The platform enables users to:

- Analyze global military rankings
- Compare two countries side-by-side
- Simulate coalition strength
- Evaluate modernization levels
- Assess defense spending intensity

Project Architecture:

Web Scrapping -> Data Cleaning -> KPI Engineering -> Dashboard Development -> QA and Final Documentation Submission.

Data Source:

Data was scraped from GlobalFirepower.com, especially from the following:

- Military Strength Ranking:
<https://www.globalfirepower.com/countries-listing.php>
- Various metric-specific pages (e.g., Total Population, Total Military Personnel, Defense Budget, etc.)

Milestone 1 – Data Collection and Preparation

Objective:

Build a reliable dataset from raw web data.

Task Completed:

- Scraped 140+ country military profiles from GlobalFirepower.com
- Extracted 50+ military indicators
- Parsed ranking, manpower, assets, and budget metrics
- Cleaned and standardized numerical values
- Handled missing data
- Converted text metrics into numeric format

Tools Used:

- Python
- Requests
- BeautifulSoup
- Pandas
- NumPy

Milestone 2 – KPI Engineering and Dashboard Preparation

Objective:

Transform raw metrics into analytical KPIs.

Engineered KPIs:

Total Aircrafts:

To measure air power, all the aircrafts types were summed up -It is performed to find out the air strength of the country

Total Ground Power:

To measure ground strength, tanks, armored vehicles and artillery is combined. -Represents the land warfare capability.

Overall Military Assets:

Combined indicator for total military strength.

Air to Ground Ratio:

This KPI shows whether a country is air dominant or ground dominant.

Military Ranks:

Countries were ranked based on the overall military assets using dense ranking.

Artillery Modernization:

To analyze modernisation the ratio of self propelled artillery to the total artillery is calculated.

Power Index Rank Gap:

Ranking in consistency using rank gap.

The Dashboard is prepared using the final files containing KPIs:

Kpi overview:

Provides the high level summary of global military strength.

- Country Slicer
- Total Ground Power
- Artillery Modernization
- Total Aircrafts
- Average Military Rank
- Ground Power Share
- Total Assets
- Total Assets Chart
- Ground Power Share (Donut Chart)

Power comparison:

Compares air and the ground strength across countries

- Country Slicer
- Total Ground Power
- Total Aircraft
- Air to Ground Ratio (Line Chart)

Modernization and ranking:

Focuses on Modernization and Ranking Consistency.

- Power index rank gap
- Artillery Modernization
- Country Slicer

Milestone 3 – Dashboard Development

Four interconnected dashboards were developed:

Quick stats and Nation Overview Dashboards-

Objective:

Provide a global snapshot of military strength rankings and trends.

Featured:

- Top 5 countries by Defense Budget
- Top 5 countries by Military Assets
- Top 5 countries by Purchasing Power Parity
- Top 5 countries by air, Naval and Land Fleet
- Power index Rank KPI Card

Nation Overview Dashboard-

Objective:

Provide detailed country-level military profile.

Features:

- Country selector
- Power Index Rank KPI Card
- Military Force Composition consisting of tanks, Aircraft and Naval
- Manpower Composition consisting of Active Personnel, Paramilitary and Reserve Personnel
- Assets Per Million Population

Compare Powers:

Objective:

Allow side-by-side comparison between any two countries.

Features:

- Dual Country Selection
- Side-by-side KPI Comparison
- Rank Gap Analysis
- Asset Category Comparison
- Budget and Manpower Comparison

Coalition Builder-

Objective:

Simulate alliance strength through multi-country aggregation.

Features:

- Coalition VS Reference Country Comparison

-Structural Composition Analysis

-Automatic Recalculation of:

- Total Assets
- Personnel
- Defense Budget

Visualization Techniques Used:

- KPI Cards
- Clustered Bar Charts
- Donut charts

Milestone 4 – Testing, QA & Deployment

Validation Performed

- KPI accuracy cross-check
- Filter & slicer testing
- Parameter validation
- Navigation flow verification
- Layout standardization
- Data aggregation checks

Final Deliverables

- Fully integrated dashboard workbook
- Structured GitHub repository
- Complete documentation
- Portfolio-ready analytics system

How to use:

1. Open the dashboard file in Power BI Desktop/ Power BI Service
2. Navigate using navigation buttons on the top.

3. Apply filters.
4. Compare countries or simulate coalitions.